Self-fulfilling fluctuations in HANK economies

Sushant Acharya and Jess Benhabib

openicpsr-243980

Overview

This replication package provides the code to replicate the analysis in our paper, using MATLAB Version 24.2.0.3070828 (R2024b) Update 7. One main file runs all the code to generate the data for the figures in the paper and outputs the value of two parameters. The replicator should expect the code to run for about 32 seconds.

Data Availability and Provenance Statements

- ☒ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

Computational requirements

Software Requirements

Matlab (code was last run with MATLAB Version 24.2.0.2863752 (R2024b)). The code was last executed on a 3.5 GHz 6-Core Intel Xeon E5 desktop with 64 GB of 1866 MHz DDR3 RAM, running MacOS version 12.7.6.

Memory and Runtime Requirements

The running time of the file is approximately 32 seconds on the desktop machine described above.

Description of programs/code

The programs provided will generate all figures in the main paper and the supplementary appendix. The file `main.m` will run them all. More specifically, `main.m` calls the following scripts:

1. `figures_for_papers.m`, which generates Figures 2a,2b, 3a, and 3b in the main paper. `figures_for_papers.m` uses (i) `makeec.m`, (ii) `model.m`, (iii) `model_phir.m` and (iv) `model_one_ss.m`.

2. `figures_for_appendix.m`, which generates Figures B.1a, B.1b,B.1c, B.1d, D.1, E.1a, E.1b, E.2a, E.2b, E.2c, E.3a, E.3b, E.4a, E.4b, E.4c in the supplementary appendix. `figures_for_appendix.m` uses (i) `colors.m`, (ii) `makeec.m`, (iii) `model.m`, (iv) `model_AMPF.m`, (v) `model_phix.m`, (vi) `model_inertial.m`, (vii) `model_changeeta.m`, (viii) `makeecHTM.m` and (ix) `eval_G.m`.

Instructions to Replicators

Run `main.m` (without adding a `-nodisplay` flag) to produce all figures in the paper and Appendix. This will save all the figures in `*.eps` format in the same folder as `main.m`.

List of tables and programs

There are no tables in the paper. The provided code reproduces:

☒ All figures in the paper (except Figure 1) and supplementary appendix, as described below. Figure 1 in the paper is qualitative and is not generated using simulated data. It was drawn manually.

Figure	Program	Line Number	Output File
Figure 2a	`figures_for_papers.m`	100	`Figure2a.eps`
Figure 2b	`figures_for_papers.m`	202	`Figure2b.eps`
Figure 3a	`figures_for_papers.m`	282	`Figure3a.eps`
Figure 3b	`figures_for_papers.m`	411	`Figure3b.eps`
Figure B.1a	`figures_for_appendix.m`	81	`Figure_B1a.eps`
Figure B.1b	`figures_for_appendix.m`	137	`Figure_B1b.eps`
Figure B.1c	`figures_for_appendix.m`	206	`Figure_B1c.eps`
Figure B.1d	`figures_for_appendix.m`	273	`Figure_B1d.eps`
Figure D.1	`figures_for_appendix.m`	319	`Figure_D1.eps`
Figure E.1a	`figures_for_appendix.m`	391	`Figure_E1a.eps`
Figure E.1b	`figures_for_appendix.m`	446	`Figure_E1b.eps`
Figure E.2a	`figures_for_appendix.m`	499	`Figure_E2a.eps`
Figure E.2b	`figures_for_appendix.m`	550	`Figure_E2b.eps`
Figure E.2c	`figures_for_appendix.m`	608	`Figure_E2c.eps`
Figure E.3a	`figures_for_appendix.m`	669	`Figure_E3a.eps`
Figure E.3b	`figures_for_appendix.m`	718	`Figure_E3b.eps`
Figure E.4a	`figures_for_appendix.m`	775	`Figure_E4a.eps`
Figure E.4b	`figures_for_appendix.m`	857	`Figure_E4b.eps`
Figure E.4c	`figures_for_appendix.m`	937	`Figure_E4c.eps`